



**BHASKARACHARYA NATIONAL INSTITUTE FOR
SPACE APPLICATIONS AND GEO-INFORMATICS**
WEEKLY PROGRESS REPORT (18/02/2025 – 24/02/2025)

WEEK 1

PROJECT NAME	Prithvi EO Models (IBM Model)
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PROJECT DESCRIPTION :

Fine tuning and Training of IBM Prithvi EO Models for Earth Observation tasks.

GROUP MEMBER :

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GROUP ID :

2026G264

GROUP GUIDE :

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COLLEGE NAME :

Gati Shakti Vishwavidyalaya, Vadodara

18/05/2026	• Learned about Prithvi EO Models, types and version of models and its functions. • Explored NASA impact Prithvi EO Github repository structure and examples
19/05/2026	• Exploring Prithvi EO models like Prithvi EO tiny, Prithvi EO 100m with sample data and what masking-based pre-training is.
20/05/2026	• Exploring various datasets which are present in the hugging face website and trained the burn_scar dataset in Prithvi 2.0 tiny tl, Prithvi 2.0 100m.
21/05/2026	• Downloaded EuroSAT and EO datasets for experimentation. • Tested basic classification workflow and verified dataset.
22/05/2026	• Downloaded Sentinel-2 tile (T42QWK) covering Gujarat for February 2024, explored real-world classification use cases for Landsat and Sentinel-2 data, finalized 4 target classes (Vegetation, Urban, Water, Barren) and understood the role of spectral bands and labelling in satellite image classification.
23/05/2026	Holiday (4th Saturday)
24/05/2026	Holiday (Sunday)

Next Week Plan	Plan to fine tune Prithvi EO models on custom datasets, test segmentation and object detection tasks, analyse model performance improvements using different training strategies.
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REFERENCE:

- [ibm-nasa-geospatial \(IBM-NASA Prithvi Models Family\)](#)
- <https://www.youtube.com/live/CB3FKtmuPI8?si=oIB4EmCtkUjSoOr>
- [Sentinel-2 | Copernicus Data Space Ecosystem](#)
- <https://research.ibm.com/blog/terramind-prithvi-tiny-small-models-geospatial>
- <https://github.com/NASA-IMPACT/Prithvi-EO-2.0>

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